



Mitoprep Premium Test 9

Date · 21 January 2026

Physics :- Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion, Gravitation

Chemistry :- Some Basic Concepts Of Chemistry, Structure of Atom, Classification of Elements & Periodicity, Chemical Bonding, Thermodynamic

Biology :- The Living World, Biological Classification, Plant Kingdom, Animal Kingdom, Morphology of Flowering Plants, Anatomy of Flowering Plant, Structural Organisation in Animals, Cell The Unit Of Life, Biomolecules, Cell Cycle & Cell Division

Instruction

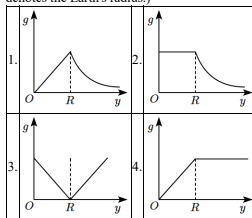
1. The test is of 3 hours duration.
2. The test booklet consists of 200 questions. The maximum mark is 720.
3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,
4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. The moment of inertia of a uniform circular lamina of mass m and radius r , about an axis tangent to the lamina (and in the plane of the lamina) is:

1. $\frac{1}{2}mr^2$	2. $\frac{3}{4}mr^2$
3. $\frac{1}{4}mr^2$	4. $\frac{5}{4}mr^2$

2. Which of the following graphs correctly represents the variation of acceleration due to gravity g as a function of the distance y from the centre of the Earth? (Here, R denotes the Earth's radius.)



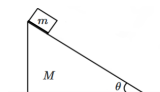
3. A small block (A) of mass 1 kg moving with a velocity of 5 m/s, strikes a stationary block (B) of mass 4 kg resting on a smooth horizontal plane. The blocks A, B stick together after the collision. The velocity of the centre-of-mass of the two blocks (A, B) before the collision is:

1. 5 m/s	2. 4 m/s
3. $\frac{5}{4}$ m/s	4. 1 m/s

4. A screw gauge has the least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is:

1. 0.25 mm	2. 0.5 mm
3. 1.0 mm	4. 0.01 mm

5. A block of mass m slides down the smooth inclined surface of a wedge of mass M starting from rest. The wedge is at rest on the horizontal surface beneath it, due to friction. The acceleration of the center of mass of the system of the block and the wedge is:



1. $\frac{mg \sin \theta}{m + M}$	2. $\frac{mg \cos \theta}{m + M}$
3. $g \sin \theta$	4. zero

6. Four identical particles of mass M are located at the corners of a square of side ' a '. What should be their speed if each of them revolves under the influence of another gravitational field in a circular orbit circumscribing the square?



1. $1.16\sqrt{\frac{GM}{a}}$	2. $1.21\sqrt{\frac{GM}{a}}$
3. $1.35\sqrt{\frac{GM}{a}}$	4. $1.41\sqrt{\frac{GM}{a}}$

7. A ball is thrown vertically downwards with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with the velocity of 80 m/s. The height of the tower is: (assuming $g = 10 \text{ m/s}^2$)

1. 340 m	2. 320 m
3. 300 m	4. 360 m

8. Three vectors $\vec{A}, \vec{B}$ and $\vec{C}$ each having magnitude of 200 units, are inclined to the x -axis at angles $45^\circ, 135^\circ$ and 315° respectively. They are added to get a resultant vector $\vec{R}$. The magnitude of $\vec{R}$ is:

1. $200\sqrt{2}$ units	2. 200 units
3. 300 units	4. zero

9. Which, of the following quantities, is dimensionally independent of mass?

1. $\frac{\text{Energy}}{\text{Time}}$	2. $\frac{\text{Energy}}{\text{Momentum}}$
3. $\text{Force} \times \text{Time}$	4. $\text{Pressure} \times \text{Time}$

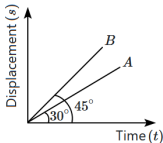
10. When a planet orbits the Sun, a force acts upon it that depends upon its orbital radius. How does the resulting tangential speed of the planet relate to the radius, assuming that the orbit is essentially circular?

1. The speed is inversely proportional to the square root of the orbital radius.
2. The speed is directly proportional to the orbital radius.
3. The speed is inversely proportional to the orbital radius.
4. The speed is inversely proportional to the square of the orbital radius.

11. A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T . If speed becomes 2ω while keeping the same radius, the tension in the string becomes:

1. $4T$	2. $\frac{T}{4}$
3. $\sqrt{2}T$	4. T

12. The displacement-time (s - t) graphs of two moving particles A and B make angles of 30° and 45° with the x -axis as shown in the figure. The ratio of their respective velocity $\left(\frac{v_A}{v_B}\right)$ is:



1. $1 : \sqrt{3}$
2. $\sqrt{3} : 1$
3. $1 : 1$
4. $1 : 2$

13. A shell of mass m is at rest initially. It explodes into three fragments having masses in the ratio $2 : 2 : 1$. If the fragments having equal masses fly off along mutually perpendicular directions with speed v , the speed of the third (lighter) fragment is:

1. $3\sqrt{2}v$	2. v
3. $\sqrt{2}v$	4. $2\sqrt{2}v$

14. On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of the same mass M which is initially at rest. After the collision, the first block moves at an angle θ to its initial direction and has a speed $\frac{v}{3}$. The second block's speed after the collision will be:

1. $\frac{2\sqrt{2}}{3}v$	2. $\frac{3}{4}v$
3. $\frac{3}{\sqrt{2}}v$	4. $\frac{\sqrt{3}}{2}v$

15. The force on a charge q in an electric field E is qE , while the force on the same charge q moving with a speed v perpendicular to a magnetic field B is qvB . Then, it can be concluded that E/B has dimensions of:

1. Electric charge
2. Electric current
3. Velocity
4. Acceleration

16. The net force acting on a projectile at the highest point of its trajectory, moving with a speed v , at a height H is:

(m is the mass of the projectile)

1. zero
2. mg
3. $mg + \frac{mv^2}{H}$
4. $mg - \frac{mv^2}{H}$

17. A projectile is launched on a horizontal surface, following a trajectory described by the equation: $y = x - \frac{x^2}{20}$, where the y -axis is directed vertically upward. The maximum height attained by the projectile is:

(all quantities are expressed in SI units)

1. 10 m
2. 5 m
3. 20 m
4. 40 m

18. Two vectors $\vec{A}$ and $\vec{B}$ are acting at an angle θ on an object. If the magnitude of their vector product is $\sqrt{3}$ times their scalar product, the angle θ between them is:

1. 30°
2. 45°
3. 60°
4. 90°

19. A particle is moving along a straight line such that its position depends on time as $x = 1 - at + bt^2$, where $a = 2 \text{ m/s}$, $b = 1 \text{ m/s}^2$. The distance covered by the particle during the first 3 seconds from start of the motion will be:

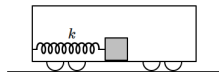
1.	2 m	2.	5 m
3.	7 m	4.	4 m

20. The magnitude of the vector

$$\hat{i} + \hat{i} \times \hat{j} + (\hat{i} \times \hat{j}) \times \hat{i} + ((\hat{i} \times \hat{j}) \times \hat{i}) \times \hat{j}:$$

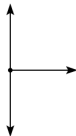
1. 1
2. $\sqrt{2}$
3. $\sqrt{3}$
4. 2

21. On the horizontal smooth floor of a railway carriage lies a spring-block system, with the spring in the relaxed state. The carriage which was travelling to the left suddenly brakes, with an acceleration equal to $\frac{g}{2}$. The maximum compression in the spring is:



1.	$\frac{mg}{k}$	2.	$\frac{mg}{2k}$
3.	$\frac{mg}{4k}$	4.	$\frac{3mg}{4k}$

22. The following diagram represents three vectors of equal magnitudes in a plane.



What arrow best represents the direction of the sum of three vectors?

1. $\rightarrow$
2. $\leftarrow$
3. $\uparrow$
4. $\downarrow$

23. Two forces of magnitude A and $\frac{A}{2}$ act perpendicular to each other. The magnitude of the resultant force is equal to:

1.	$\frac{A}{2}$	2.	$\frac{\sqrt{5}A}{2}$
3.	$\frac{3A}{2}$	4.	$\frac{5A}{2}$

24. A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration, if the kinetic energy of the particle becomes equal to 8×10^{-4} J by the end of the second revolution after the beginning of the motion?

1. 0.15 m/s^2	2. 0.18 m/s^2
3. 0.2 m/s^2	4. 0.1 m/s^2

25. A block of mass m is attached to an unstretched spring with a spring constant k placed on a smooth horizontal table. The block is now pulled to a displacement of x_m and then released. The maximum speed it will attain will be:

1. $\sqrt{\frac{k}{m}} x_m$	2. $\sqrt{\frac{m}{k}} x_m$
3. $\frac{k}{m} x_m$	4. $\frac{m}{k} x_m$

26. If force (F), velocity (v), and time (T) are taken as fundamental units, the dimensions of mass will be:

1. $[FvT^{-1}]$	2. $[FvT^{-2}]$
3. $[Fv^{-1}T^{-1}]$	4. $[Fv^{-1}T]$

27. What is the depth at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times its value at the surface of the earth? (radius of the earth = R)

1. $\frac{R}{n^2}$	2. $\frac{R(n-1)}{n}$
3. $\frac{Rn}{(n-1)}$	4. $\frac{R}{n}$

28. The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is:

- $\theta = \tan^{-1}\left(\frac{1}{4}\right)$
- $\theta = \tan^{-1}(4)$
- $\theta = \tan^{-1}(2)$
- $\theta = 45^\circ$

29. A cricket ball is thrown with a speed of 32 ms^{-1} at an angle of 30° above the horizontal. The maximum height it will attain is: (take $g = 10 \text{ ms}^{-2}$)

- 0.8 m
- 12.8 m
- 25.6 m
- 51.2 m

30. A man walks on the ground (horizontal) with a steady acceleration a . The coefficient of static friction μ must satisfy:

- $\mu \leq \frac{a}{g}$
- $\mu \leq \frac{g}{a}$
- $\mu \geq \frac{a}{g}$
- $\mu \geq \frac{g}{a}$

31. Given below are two statements:

Assertion (A):	If the same mass is drawn into the making of a solid sphere and a hollow sphere, the moment of inertia about the diameter of a hollow body is greater than that of a solid body.
Reason (R):	The moment of inertia depends on the mass distribution from the axis of rotation and in a hollow sphere, more of the mass is further away from the axis of rotation.

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	Both (A) and (R) are False.

32. Given below are two statements:

Assertion (A):	The centre-of-mass of a solid body lies within the body.
Reason (R):	The centre-of-mass of any two particles lies between them, on the straight line segment joining them.

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	(A) is False but (R) is True.

33. During uniformly accelerated motion in a straight line, which of the following can be negative or positive?

1. distance travelled	2. time travelled
3. average velocity	4. average speed

34. A ball is thrown at an angle θ_0 above the horizontal, and follows the parabolic path taken by a projectile. Let its speed be v when its trajectory makes an angle θ with the horizontal. Assuming A to be a constant,

- $v = A \cos \theta$
- $v = A \sin \theta$
- $v = A \tan \theta$
- $v = A \sec \theta$

35. The main scale of a vernier caliper has the least count of 1 mm. 20 divisions of the vernier scale coinciding with 19 divisions of the main scale. The vernier constant of the caliper is:

- 0.01 cm
- 0.01 mm
- 0.005 cm
- 0.005 mm

36. An electric lift with a maximum load of 2000 kg (lift+passengers) is moving up with a constant speed of 1.5 ms^{-1} . The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is:

(Take $g = 10 \text{ ms}^{-2}$)

- 23500
- 23000
- 20000
- 34500

37. A small block (A) of mass 1 kg moving with a velocity of 5 m/s, strikes a stationary block (B) of mass 4 kg resting on a smooth horizontal plane. The blocks A, B stick together after the collision.

The loss in kinetic energy of the blocks in the collision is:

1.	10 J	2.	20 J
3.	12.5 J	4.	15 J

38. A planet of mass m , moves in a circular orbit around the sun (mass: $M, M \gg m$); the angular momentum of the planet in its orbit being L . The speed of the planet in its orbit equals:

1.	$\frac{GMm}{2L}$	2.	$\frac{GMm}{L}$
3.	$\frac{2GMm}{L}$	4.	$\frac{4GMm}{L}$

39. Given below are two statements:

Assertion (A):	On a rainy day, it is difficult to drive a car or a bus at a high speed.
Reason (R):	The value of the coefficient of friction is lowered when the surface is wetted.

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	Both (A) and (R) are False.

40. A machine gun is firing 10 g bullets with a speed of 250 m/s. To keep the machine gun in its position, 125 N force is required. The number of bullets fired per second are:

1.	40	2.	50
3.	25	4.	10

41. Imagine that there exists a planet whose mass and radius are both half that of the Earth. The acceleration due to gravity on that planet will be:

- $g/4$
- $g/2$
- $2g$
- $4g$

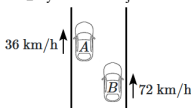
42. The energy that will be ideally radiated by a 100 kW transmitter in 1 hr is:

1. $1 \times 10^5 \text{ J}$	2. $36 \times 10^7 \text{ J}$
3. $36 \times 10^4 \text{ J}$	4. $36 \times 10^5 \text{ J}$

43. A block of mass 5 units is moving horizontally at speed of 1.5 ms^{-1} . A vertically upward force 5 N acts on it for 4 s. What will be the distance of the block from the point where the force starts acting after 4 s? (Ignore gravity)

- 2 m
- 6 m
- 14 m
- 10 m

44. Two cars A and B, each 5 m long, are travelling along parallel lanes with speeds of 36 km/hr and 72 km/hr. The car B is just behind the car A, but is beginning to overtake. What will be the distance covered by the car B by the time it is just ahead of the car A?



- 10 m
- 15 m
- 20 m
- 25 m

45. A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of the table and from its other end, another block B of mass m_2 is suspended. The coefficient of kinetic friction between block A and the table is μ_k . When block A is sliding on the table, the tension in the string is:

1. $\frac{(m_2 + \mu_k m_1)g}{(m_1 + m_2)}$	2. $\frac{(m_2 - \mu_k m_1)g}{(m_1 + m_2)}$
3. $\frac{m_1 m_2 (1 - \mu_k)g}{(m_1 + m_2)}$	4. $\frac{m_1 m_2 (1 + \mu_k)g}{m_1 + m_2}$

Chemistry

46. Given:

(i)	$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ $\Delta H^\circ_{298 \text{ K}} = -285.9 \text{ kJ mol}^{-1}$
(ii)	$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$ $\Delta H^\circ_{298 \text{ K}} = -241.8 \text{ kJ mol}^{-1}$

The molar enthalpy of vapourisation of water will be:

1. 74.8 kJ mol^{-1}	2. 22.0 kJ mol^{-1}
3. 44.1 kJ mol^{-1}	4. 52.7 kJ mol^{-1}

47. The correct shape and hybridisation of BrF_5 is:

- Square pyramidal and sp^3d^2
- Square pyramidal and d^2sp^3
- Trigonal bipyramidal and sp^3d^2
- Trigonal bipyramidal and d^2sp^3

48. Which of the following compounds does not possess a dipole moment?

- CH_3Cl
- H_2O
- C_6H_6 (Benzene)
- CH_2N_2

49. Which among the following are isoelectronic species?

- Na^+ , K^+ , Ca^{2+} , Mg^{2+}
- Ca^{2+} , Mg^{2+} , S^{2-} , K^+
- Ca^{2+} , Ar , S^{2-} , K^+
- None of the above

50. Match Column I (Reactions) with Column II (Enthalpy) and mark the appropriate choice:

Column I (Reactions)	Column II (Enthalpy)
(A) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	(i) $\Delta_{\text{sol}}H^\circ$
(B) $\text{H}_2(\text{g}) \rightarrow 2\text{H}(\text{g})$	(ii) $\Delta_{\text{lattice}}H^\circ$
(C) $\text{NaCl}(\text{s}) \rightarrow \text{Na}^+_{(\text{g})} + \text{Cl}^-_{(\text{g})}$	(iii) $\Delta_{\text{c}}H^\circ$
(D) $\text{NaCl}(\text{s}) \rightarrow \text{Na}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$	(iv) $\Delta_{\text{bond}}H^\circ$

- (A) $\rightarrow$ (iv), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (i), (D) $\rightarrow$ (ii)
- (A) $\rightarrow$ (ii), (B) $\rightarrow$ (i), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (iii)
- (A) $\rightarrow$ (i), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (iv)
- (A) $\rightarrow$ (iii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (i)

51. Which of the following is the most basic oxide?

1. Al_2O_3
2. Sb_2O_3
3. Bi_2O_3
4. SeO_2

52. Match the items of Column I with items of Column II.

Column I (Physical quantity)	Column II (Unit)
A. Molarity	1. mol kg^{-1}
B. Mole fraction	2. mol
C. Mole	3. mol L^{-1}
D. Molality	4. Unitless

Choose the correct match:

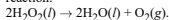
1. A-3, B-4, C-2, D-1
2. A-1, B-4, C-2, D-3
3. A-2, B-4, C-3, D-2
4. A-2, B-3, C-4, D-1

53. The energy of an electron is given by $-2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$. Find out the wavelength of light required to excite an electron in a hydrogen atom from level $n = 1$ to $n = 2$:

(Given: $h = 6.62 \times 10^{-34} \text{ Js}$ and $c = 3.0 \times 10^8 \text{ ms}^{-1}$)

1. $1.214 \times 10^{-7} \text{ m}$
2. $2.816 \times 10^{-10} \text{ m}$
3. $6.500 \times 10^{-17} \text{ m}$
4. $8.500 \times 10^{-13} \text{ m}$

54. Determine the enthalpy change for the specified reaction:



(Given the heat of formation of $\text{H}_2\text{O}_2(l)$ and $\text{H}_2\text{O}(l)$ are -188 and -286 kJ/mol respectively)

1. -196 kJ/mol	2. $+196 \text{ kJ/mol}$
3. $+948 \text{ kJ/mol}$	4. -948 kJ/mol

55. Which of the following equation represents a reaction that provides the enthalpy of the formation of CH_3Cl ?

1. $\text{C}(s) + \text{HCl}(g) + \text{H}_2(g) \rightarrow \text{CH}_3\text{Cl}(g)$
2. $\text{C}(s) + 3\text{H}(g) + \text{Cl}(g) \rightarrow \text{CH}_3\text{Cl}(g)$
3. $\text{C}(s) + 3/2\text{H}_2(g) + 1/2\text{Cl}_2(g) \rightarrow \text{CH}_3\text{Cl}(g)$
4. $\text{CH}_4(g) + \text{Cl}_2(g) \rightarrow \text{CH}_3\text{Cl}(g) + \text{HCl}(g)$

56. Of the following transitions in hydrogen atom, the one which gives an absorption line of lowest frequency is:

1. $n = 1$ to $n = 2$
2. $n = 3$ to $n = 8$
3. $n = 2$ to $n = 1$
4. $n = 8$ to $n = 3$

57. The number of d-electrons in Fe^{2+} (atomic number $Z = 26$) is different from the number of:

1. s-electrons in Mg ($Z = 12$)
2. p-electrons in Cl ($Z = 17$)
3. d-electrons in Fe ($Z = 26$)
4. p-electrons in Ne ($Z = 10$)

58. The amount of CO_2 that can be produced when 2 moles of carbon are burnt in 16 g of dioxygen is:

1. 22 g
2. 44 g
3. 0.5 g
4. 11 g

59. Consider the given two statements:

Assertion (A):	The first ionization enthalpy of aluminum is lower than that of magnesium.
Reason (R):	Ionic radius of aluminum cation (Al^{3+}) is smaller than that of magnesium cation (Mg^{2+}).

1.	Both (A) and (R) are True and (R) correctly explains (A)
2.	Both (A) and (R) are True but (R) does not correctly explain (A)
3.	(A) is True but (R) is False
4.	Both (A) and (R) are False

60. Heat at constant volume and heat at constant pressure are, respectively:

1. ΔG and ΔH
2. ΔH and ΔE
3. ΔE and ΔH
4. ΔE and ΔG

61. Match the electronic configurations in List-I with their corresponding ionization energies in List-II.

List-I (Electronic configuration of elements)	List-II ($(I.E)_1$ in kJ mol^{-1})
a. $1s^2 2s^2$	(i) 801
b. $1s^2 2s^2 2p^4$	(ii) 899
c. $1s^2 2s^2 2p^3$	(iii) 1314
d. $1s^2 2s^2 2p^1$	(iv) 1402

Choose the most appropriate answer from the options given below:

1. (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)
2. (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)
3. (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (ii)
4. (a) $\rightarrow$ (iv), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iii)

62. Two electrons occupying the same orbital are distinguished by:

1. Magnetic quantum number
2. Azimuthal quantum number
3. Spin quantum number
4. Principal quantum number

63. Which of the following pair is correctly matched?

1. Basic oxides - In_2O_3 , K_2O , SnO_2
2. Neutral oxides - CO , NO_2 , N_2O
3. Acidic oxides - Mn_2O_7 , SO_2 , TeO_3
4. Amphoteric oxides - BeO , Ga_2O_3 , GeO

64. Three hydrochloric acid solutions, X, Y, and Z, with molarities of 0.07 M, 0.12 M, and 0.15 M, respectively, are mixed to prepare 100 mL of a 0.1 M solution.

What volumes of X, Y, and Z should be used?

1. 50 mL, 25 mL, 25 mL
2. 20 mL, 60 mL, 20 mL
3. 40 mL, 30 mL, 30 mL
4. 55 mL, 20 mL, 25 mL

65. Match List I with List II.

List I Quantum Number	List II Information provided
A. m_l	I. Shape of the orbital
B. m_s	II. Size of the orbital
C. l	III. Orientation of the orbital
D. n	IV. Orientation of spin of the electron

Choose the correct answer from the options given below:

1. A-III, B-IV, C-I, D-II
2. A-III, B-IV, C-II, D-I
3. A-II, B-I, C-IV, D-III
4. A-I, B-III, C-II, D-IV

66. How many significant figures are present in the following numbers?

(I)	161 cm
(II)	0.0161
(III)	1.61

1. 3, 3, 3
2. 3, 4, 3
3. 3, 2, 3
4. 3, 4, 4

67. Which of the following pairs of ions are isoelectronic and isostructural?

1. CO_3^{2-} , SO_3^{2-}
2. ClO_3^- , CO_3^{2-}
3. SO_3^{2-} , NO_3^-
4. ClO_3^- , SO_3^{2-}

68. How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms?

1. 0.02
2. 3.125×10^{-2}
3. 1.25×10^{-2}
4. 2.5×10^{-2}

69. According to the periodic law of elements, the variation in properties of elements is related to their:

1. Atomic masses
2. Nuclear mass
3. Atomic numbers
4. Nuclear neutron-proton number ratio

70. The bond order of C_2 molecule is two. Therefore, it can be concluded that it has a double bond. Consequently, it can also be concluded that C_2 molecule contains:

1. Two sigma bonds
2. Two pi bonds
3. One sigma and one pi bond
4. Half sigma and one and half pi bond

71. A pair of d-orbitals having electron density along the axes is:

1. d_{xz} , d_{yz}
2. d_{xz} , d_{xy}
3. d_{xz} , $d_{x^2-y^2}$
4. d_{xy} , $d_{x^2-y^2}$

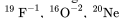
72. 1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186 moles of CO_2 . The molar mass of M_2CO_3 in $g\ mol^{-1}$ is:

1. 118.6
2. 11.86
3. 88.6
4. 84.3

73. Which of the following statements is correct for an element that forms oxides in different oxidation states?

- | | |
|----|--|
| 1. | The oxide is neutral when the element is in its highest oxidation state. |
| 2. | The oxide is most acidic when the element is in its highest oxidation state. |
| 3. | The oxide is amphoteric when the element is in its highest oxidation state. |
| 4. | The oxide is most basic when the element is in its highest oxidation state. |

74. Choose the correct statement considering the information given below.



1. Both O^{-2} and F^{-} are isoelectronic.
2. All given ions have an equal number of e^{-}
3. F^{-} and Ne have an equal number of e^{-}
4. All are correct.

75. If water vapour is assumed to be a perfect gas, molar enthalpy change for vapourisation of 1 mol of water at 1 bar and $100^{\circ}C$ is $41\ kJ\ mol^{-1}$. The internal energy change, when 1 mol of water is vapourised at 1 bar pressure and $100^{\circ}C$ is:

1. $35.5\ kJ\ mol^{-1}$	2. $37.9\ kJ\ mol^{-1}$
3. $41\ kJ\ mol^{-1}$	4. $44.2\ kJ\ mol^{-1}$

76. Considering the data given below, the radius of the second Bohr orbit for a hydrogen atom is:

(Planck's constant $h = 6.6262 \times 10^{-34}\ J\ s$; mass of electron = $9.1091 \times 10^{-31}\ kg$;

charge of electron $e = 1.60210 \times 10^{-19}\ C$)

1. $1.65\ \text{\AA}$
2. $4.76\ \text{\AA}$
3. $0.529\ \text{\AA}$
4. $2.12\ \text{\AA}$

77. A sample of a pure compound was found to contain carbon, hydrogen, and chlorine.

Its elemental analysis is given below:

Element	Composition/Weight
Carbon	1.201 grams
Hydrogen	0.202 grams
Chlorine	7.090 grams

What is the empirical formula for the compound?

1. $CHCl_3$	2. CH_2Cl
3. CH_2Cl_2	4. CH_3Cl

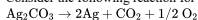
78. When 1.0 g of magnesium (Mg) is reacted with 0.56 g of oxygen (O_2) in a closed container, determine the reactant left in excess and calculate the mass of the excess reactant remaining after the reaction is complete: (At. wt. $Mg = 24$; $O = 16$)

1. Mg , 0.16 g	2. O_2 , 0.16 g
3. Mg , 0.44 g	4. O_2 , 0.28 g

79. Which of the following species has a bond angle of 120° ?

1. SF_6
2. NCl_3
3. BCl_3
4. PH_3

80. Upon strong heating, what is the weight of the residue produced by 2.76 gram silver carbonate? Consider the following reaction for reference:



1. 2.16 g
2. 2.48 g
3. 2.32 g
4. 2.64 g

81. A sample set of electrons have the following quantum numbers:

(i)	$n = 4, l = 1$
(ii)	$n = 4, l = 0$
(iii)	$n = 3, l = 2$
(iv)	$n = 3, l = 1$

Arrange them in the increasing order of energy from lowest to highest:

1.	$(iv) < (ii) < (iii) < (i)$
2.	$(ii) < (iv) < (i) < (iii)$
3.	$(i) < (iii) < (ii) < (iv)$
4.	$(iii) < (i) < (iv) < (ii)$

82. Which of the following molecules contain bond angles of 180° in their gaseous states?

(Atomic numbers: H = 1; Be = 4; C = 6; Cl = 17)

- A. BeCl_2
B. C_2H_2
C. CCl_4

1. A and C only	2. A and B only
3. A only	4. B and C only

83. Which of the following processes is endothermic?

1. $\text{O} + e^- \rightarrow \text{O}^-$
2. $\text{N} + e^- \rightarrow \text{N}^-$
3. $\text{Cl} + e^- \rightarrow \text{Cl}^-$
4. None of the above

84. Which of the following is/are correct about isothermal expansion of work?

1. $w_{\text{irreversible}} = w_{\text{reversible}}$
2. $w_{\text{irreversible}} > w_{\text{reversible}}$
3. $w_{\text{irreversible}} < w_{\text{reversible}}$
4. All of the above

85. Which of the following represents the correct ground-state electronic configuration of a gas-phase Mn^{2+} ion?

1. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$
2. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
3. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$
4. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$

86. For which of the following reaction, $\Delta H < \Delta E$?

1.	$\text{N}_2(g) + 3\text{H}_2(g) \rightarrow 2\text{NH}_3(g)$
2.	$\text{PCl}_5(g) \rightarrow \text{PCl}_3(g) + \text{Cl}_2(g)$
3.	$\text{H}_2(g) + \text{I}_2(g) \rightarrow 2\text{HI}(g)$
4.	$\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$

87. In which of the following reactions is the enthalpy change equal to the enthalpy of formation?

1. C (Graphite) + $\frac{1}{2}\text{O}_2(g) \rightarrow \text{CO}(g)$
2. S (Rhombic) + $\text{O}_2(l) \rightarrow \text{SO}_2(g)$
3. 4P (White) + $3\text{O}_2(g) \rightarrow 2\text{P}_2\text{O}_3(s)$
4. $\text{H}_2(g) + \text{I}_2(g) \rightarrow 2\text{HI}(g)$

88. Match the periodic properties given in List I with the corresponding elements given in List II:

List I	List II (High value observed for)
A. Ionisation potential	I. Chlorine
B. Electropositivity	II. Caesium
C. Electron affinity	III. Helium
D. Oxidation state	IV. Fluorine
	V. Osmium

Choose the correct answer from the options given below:

1. A-IV, B-III, C-II, D-I
2. A-III, B-II, C-I, D-V
3. A-I, B-II, C-III, D-IV
4. A-II, B-I, C-IV, D-V

89. Anions are larger in radius than their parent atoms because:

1.	An anion has a fewer number of electrons than its parent atom
2.	An anion has the same number of electrons as its parent atom.
3.	An anion has a higher effective nuclear charge.
4.	An anion has more electrons than its parent atom

90. The correct shape and hybridization for XeF_4 are:

1. octahedral, sp^3d^2
2. trigonal bipyramidal, sp^3d^3
3. planar triangle, sp^3d^3
4. square planar, sp^3d^2

Biology

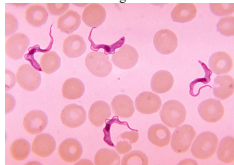
91. In *Selaginella* and *Equisetum*:

1.	spores produced are of one kind.
2.	heterospory is seen.
3.	gametophyte is the dominant generation in the life cycle.
4.	sporophylls form distinct compact structures called strobili or cones.

92. Casparian strips occur in?

1. Epidermis	2. Pericycle
3. Cortex	4. Endodermis

93. The blood film shows the flagellated organism responsible for causing African sleeping sickness in humans. Name the organism:



1. *Clonorchis sinensis*
2. *Brugia malayi*
3. *Trypanosoma brucei*
4. *Ancylostoma duodenale*

94. Consider the given two statements:

I:	Meiosis preserves conservation of specific chromosome number of each species across generations in sexually reproducing organisms.
II:	Meiosis increases the genetic variability in the population of organisms from one generation to the next.

1. Only **I**
2. Only **II**
3. Both **I** and **II** are correct
4. Neither **I** nor **II**

95. An analogous variant of the agent that causes BSE in cattle causes:

1. Potato spindle tuber disease
2. Cr-Jacob disease (CJD) in humans
3. Mosaic formation in plants
4. Leaf rolling and curling in plants

96. The sporophyte of mosses is more elaborate than that in:

1. Pteridophytes	2. Gymnosperms
3. Angiosperms	4. Liverworts

97. During prophase I of meiosis I, recombination between homologous chromosomes is completed:

1. in the Zygotene
2. in the beginning of the Pachytene
3. by the end of Pachytene
4. in the beginning of the Diplotene

98. The cell shown in the given figure is most likely at:



1. Metaphase I
2. Metaphase II
3. Anaphase I
4. Anaphase II

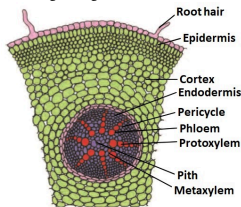
99. Match List-I with List-II:

List-I (Biological Name)	List-II (Common name)
A. <i>Asterias</i>	I. Sea urchin
B. <i>Antedon</i>	II. Brittle star
C. <i>Echinus</i>	III. Sea lily
D. <i>Ophiura</i>	IV. Star fish

Choose the correct answer from the options given below:

1. A-III, B-IV, C-II, D-I
2. A-IV, B-III, C-I, D-II
3. A-II, B-III, C-I, D-IV
4. A-I, B-II, C-IV, D-III

100. The given figure shows a T.S. of:



1. Monocot root	2. Dicot root
3. Dicot stem	4. Monocot stem

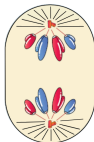
101. Match each item in Column I with one in Column II and select the correct match from the codes given:

Column I [Organ]	Column II [Lining epithelium]
A PCT	P Simple squamous
B Small intestine	Q Ciliated columnar
C Alveoli	R Columnar brush bordered
D Fallopian tube	S Cuboidal brush bordered

Codes:

	A	B	C	D
1.	R	S	P	Q
2.	R	S	Q	P
3.	S	R	P	Q
4.	S	R	Q	P

102. Identify the stage of mitosis shown in the given figure:



1. Prophase	2. Metaphase
3. Anaphase	4. Telophase

103. Biflagellate zoospores that are pear-shaped and have two unequal laterally attached flagella are involved in:

1. Asexual reproduction in brown algae
2. Asexual reproduction in red algae
3. Sexual reproduction in brown algae
4. Sexual reproduction in red algae

104. How many of the given statements are true?

A:	Alternate phyllotaxy is seen in China Rose, Mustard, and Sunflower.
B:	Opposite phyllotaxy is seen in Calotropis and Guava.
C:	Whorled phyllotaxy is seen in Alstonia.

Mark the correct option:

- 0
- 1
- 2
- 3

105. Match each item in Column I with one in Column II and choose the correct option from the codes given below:

Column I	Column II
A. Psilopsida	P. Selaginella
B. Lycopsida	Q. <i>Adiantum</i>
C. Sphenopsida	R. <i>Equisetum</i>
D. Pteropsida	S. <i>Psilotum</i>

	A	B	C	D
1.	R	Q	S	P
2.	S	P	R	Q
3.	P	S	Q	R
4.	Q	R	P	S

106. What is a polysome?

1.	Several messenger RNA molecules attached to the larger subunit of prokaryotic ribosome.
2.	Several ribosomes attached to a single messenger RNA molecule.
3.	Three major types of cellular RNAs, namely mRNA, rRNA, and tRNA, all attached to a single messenger RNA molecule.
4.	Many ribosomes attached to the surface of endoplasmic reticulum.

107. What would be the best description of stomata?

1.	two guard cells plus the subsidiary cells only.
2.	an opening surrounded by two guard cells.
3.	an opening surrounded by two guard cells and the subsidiary cells.
4.	two guard cells only.

108. What type of coloration allows frogs to change color to hide from predators?

- Cryptic coloration
- Aposematic coloration
- Camouflage
- Disruptive coloration

109. Select the *wrong* statement:

1.	Cell wall is present in members of Fungi and Plantae.
2.	Mushrooms belong to Basidiomycetes.
3.	Pseudopodia are locomotory and feeding structures in Sporozoans.
4.	Mitochondria are the powerhouse of the cell in all kingdoms except Monera.

110. Systematics does not include-

- Identification
- Nomenclature
- Classification
- Evolutionary relationships between environment and organisms

111. How many of the given statements regarding enzymes are correct?

I:	Most are proteins.
II:	They increase the requirement for activation energy for a biochemical reaction.
III:	They are used up in the biochemical reactions they catalyze.

- None
- Only **I**
- Only **II**
- All **III**

112. The suffix that denotes the taxonomic category 'Family' in plants would be

- ae
- aceae
- ales
- nae

113. In Whittaker's five Kingdom classification, organisms with eukaryotic cell type and cellular body organisation are kept under:

1. Kingdom Monera	2. Kingdom Protista
3. Kingdom Fungi	4. Kingdom Plantae

114. Regarding factors affecting enzyme activity:

I:	Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys enzymatic activity.
II:	With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first.

1.	Only I is correct
2.	Only II is correct
3.	Both I and II are correct
4.	Both I and II are incorrect

115. The "active site" of an enzyme is:

1.	The sequence of amino acids that forms the enzyme's backbone.
2.	The crevice or pocket where the substrate binds.
3.	A part of the enzyme that gets destroyed at low temperatures.
4.	A non-functional region involved in structural stability.

116. Sapindales is which taxon?

1.	Order	2.	Class
3.	Genus	4.	Species

117. Select the option that does not represent the taxa at different levels:

1.	Chordate, Mammal, Dog
2.	Animal, Insect, Butterfly
3.	Plant, Monocot, Wheat
4.	Mammal, Insect, Dicot

118. The members of deuteromycetes reproduce only by asexual spores called as:

1.	Zoospores	2.	Sporangiospores
3.	Aplanospores	4.	Conidia

119. When the co-factor is removed from the enzyme:

1. catalytic activity is lost
2. catalytic activity is reduced
3. catalytic activity increases
4. catalytic activity remains same

120. How many of the given statements regarding cell wall are true?

I:	A non-living rigid structure called the cell wall forms an outer covering for the plasma membrane of fungi and plants.
II:	Cell wall not only gives shape to the cell and protects the cell from mechanical damage and infection, it also helps in cell-to-cell interaction and provides barrier to undesirable macromolecules.
III:	Algae have cell wall, made of cellulose, galactans, mannans and minerals like calcium carbonate, while in other plants it consists of cellulose, hemicellulose, pectins and proteins.
IV:	The cell wall of a young plant cell, the primary wall is capable of growth.
V:	The secondary wall is formed on the outer (towards middle lamella) side of the cell.

1. 2
2. 3
3. 4
4. 5

121. Scientific name does not ensure

1.	Each organism has only one name
2.	Description of any organism lead to the same name of organism in any part of the world
3.	No two organisms have the same name
4.	Status of threat of extinction of that organism holding a specific scientific name.

122. Which group of animals belong to the same phylum?

1.	Portuguese men-of-war, sea anemone, and brain coral.
2.	Tapeworm, roundworm, and hookworm.
3.	Pearl oyster, devil fish, and flying fish.
4.	Sea urchin, sea cucumber, and sea pen.

123. How many of the given statements regarding frog are correct?

A:	The frog never drinks water but absorbs it through the skin.
B:	Frogs respire through skin when in water.
C:	The vascular system of frog is well-developed closed type.
D:	In male frogs, the ureters act as urogenital duct.
E:	The ear is an organ of hearing as well as balancing.

1. 2
2. 3
3. 4
4. 5

124. Which one of the following categories of animals, is correctly described with no single exception in it?

1.	All reptiles possess scales, have a three chambered heart and are cold blooded (poikilothermal).
2.	All bony fishes have four pairs of gills and an operculum on each side.
3.	All sponges are marine and have collared cells.
4.	All mammals are viviparous and possess diaphragm for breathing.

125. Which of the following will not be true regarding the floral characters of members of Solanaceae?

1.	Flowers are bisexual and zygomorphic.
2.	Calyx has five sepals which are united, persistent with velvet pubescence.
3.	Androecium has five stamens that are epipetalous.
4.	Gynoecium has bicarpellary, obliquely-placed syncarpous superior ovary.

126. What is the primary structural precursor to the new cell wall formed during plant cytokinesis?

1. Cell plate
2. Middle lamella
3. Spindle fibres
4. Metaphase plate

127. Rust and smut fungi belong to the fungal class:

1. Ascomycetes	2. Phycmycetes
3. Deuteromycetes	4. Basidiomycetes

128. Consider the two statements:

Statement I:	Green algae are usually grass green due to the dominance of pigments chlorophyll a and b.
Statement II:	Red algae vary in colour from olive green to various shades of brown depending upon the amount of fucoxanthin.

1. **Statement I** is correct; **Statement II** is correct
2. **Statement I** is incorrect; **Statement II** is correct
3. **Statement I** is correct; **Statement II** is incorrect
4. **Statement I** is incorrect; **Statement II** is incorrect

129. Petiole expands, becomes green and synthesizes food in:

1. *Opuntia*
2. *Euphorbia*
3. *Australian acacia*
4. *Colocasia*

130. All the following will be correct for bryophytes in general, except:

1.	The sex organs are multicellular.
2.	The female sex organ called archegonium is flask-shaped and produces a single egg.
3.	They produce biflagellate antherozoids.
4.	Zygotes undergo reduction division immediately.

131. Consider the two statements:

Statement I:	Bryophytes usually occur in damp, humid and shaded localities and are not truly 'successful' on land.
Statement II:	They are dependent on water for sexual reproduction.

1. **Statement I** is correct and **Statement II** is incorrect
2. Both **Statement I** and **Statement II** are correct and **Statement II** explains **Statement I**
3. **Statement I** is incorrect and **Statement II** is correct
4. Both **Statement I** and **Statement II** are correct but **Statement II** does not explain **Statement I**

132. Match List-I with List-II.

List-I	List-II
A. F ₁ Particles	I. Chromosomes
B. Histones	II. Cilia
C. Axoneme	III. Golgi apparatus
D. Cisternae	IV. Mitochondria

Choose the correct answer from the options given below:

1. A-II, B-I, C-IV, D-III
2. A-IV, B-I, C-II, D-III
3. A-IV, B-I, C-III, D-II
4. A-IV, B-III, C-I, D-II

133. In most eubacterial cells, the cell envelope consists of a tightly bound three layered structure where the outermost is the:

1. Middle lamella
2. Glycocalyx
3. Cell wall
4. Cell membrane

134. Which of the following is not a component of a nucleotide?

1. Phosphate group
2. Ribose sugar
3. Fatty acid
4. Nitrogenous base

135. The true macromolecular fraction of any living tissue or cell includes all the following biomolecules except:

1. Nucleic acids	2. Polysaccharides
3. Lipids	4. Proteins

136. Which of the following statements regarding epithelial tissue in different organs is correct?

A.	Microvilli in the proximal convoluted tubule (PCT) of the nephron increase surface area for filtration.
B.	Simple columnar epithelium is composed of multiple layers of tall, slender cells with nuclei at the base, primarily aiding in protection.
C.	Ciliated epithelium is found lining the bronchioles and fallopian tubes, where it moves particles or mucus in a specific direction.
D.	Cuboidal epithelium in the PCT of the nephron mainly facilitates mucus secretion rather than absorption.

Choose the correct option:

1. Only A and C are correct
2. Only B and D are correct
3. Only C is correct
4. Only A, C and D are correct

137. One full turn of the double stranded B-DNA helical strand would involve:

1. four base pairs
2. ten base pairs.
3. 20 base pairs
4. 200 base pairs

138. Skeletal muscles are attached to bones by structures called:

1. Cartilages	2. Ligaments
3. Tendons	4. Fasciae

139. Consider the following statements about the stages of prophase I in meiosis:

(i)	Leptotene stage is marked by the gradual visibility of chromosomes under the light microscope.
(ii)	Zygotene stage involves the pairing of homologous chromosomes, forming synaptonemal complexes.
(iii)	Pachytene stage is where the crossing over between non-sister chromatids occurs.

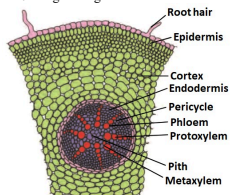
Which of the above statements are correct?

1. Only (i) and (ii)
2. Only (ii) and (iii)
3. Only (i) and (iii)
4. All (i), (ii) and (iii)

140. Example of an edible underground stem is:

1. Carrot
2. Groundnut
3. Sweet potato
4. Potato

141. The given figure shows the T.S. of a:



1. Monocot root
2. Dicot root
3. Dicot stem
4. Monocot stem

142. Which of the following statements about numerical taxonomy is incorrect?

1.	It is based on the quantification of observable characteristics and the use of statistical methods.
2.	It gives equal importance to all characteristics and does not distinguish between homologous and analogous traits.
3.	It relies solely on genetic data and molecular markers for classification.
4.	It uses computer algorithms to analyze large datasets and determine relationships between organisms.

143. Match the animal classes in Column I with their primary characteristics in Column II:

	Column I		Column II
A	Reptilia	P	Egg spawning on water surface
B	Amphibia	Q	Presence of mammary glands
C	Aves	R	Skin covered with scales or scutes
D	Mammalia	S	Endothermic with feathers

Codes:

	A	B	C	D
1.	R	P	Q	S
2.	Q	P	R	S
3.	R	P	S	Q
4.	S	P	Q	R

144. All the statements regarding comparison of prokaryotic cells and eukaryotic cells are correct except:

1.	A semi-fluid matrix called cytoplasm occupies the volume of both prokaryotic and eukaryotic cell.
2.	Cells that have membrane bound nuclei are called eukaryotic whereas cells that lack a membrane bound nucleus are prokaryotic.
3.	The prokaryotic cells lack membrane bound organelles such as those that are present in eukaryotic cells.
4.	Ribosomes are double membrane bound organelles in eukaryotic cells and non-membrane bound organelles in prokaryotic cells.

145. Simultaneous splitting of centromeres of each chromosome during mitosis takes place during:

1. Prophase	2. Metaphase
3. Anaphase	4. Telophase

146. Which of the following features will be seen in chordates but not in non-chordates?

(a)	A notochord
(b)	Ventral, solid, and double central nervous system
(c)	Pharynx perforated by gill slits
(d)	Ventral heart
(e)	Terminal anus and a lack of post-anal tail

1. (a), (c), and (d) only	2. (a), (b) and (e) only
3. (a) and (c) only	4. (b), (c) and (d) only

147. Consider the given two statements:

Statement I:	Male frogs can be distinguished by the presence of vocal sacs and a copulatory pad.
Statement II:	Female frogs have vocal sacs and copulatory pads.

- Both **Statement I** and **Statement II** are correct.
- Both **Statement I** and **Statement II** are incorrect.
- Statement I** is correct but **Statement II** is incorrect.
- Statement I** is incorrect but **Statement II** is correct.

148. Consider the given two statements:

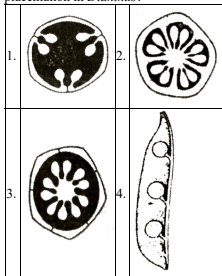
Assertion (A):	Collenchyma cells provide structural support to growing parts of the plant such as young stems and petioles.
Reason (R):	Collenchyma cells have thickened secondary cell walls that are lignified.

1. Both (A) and (R) are True and (R) correctly explains (A)
2. Both (A) and (R) are True but (R) does not correctly explain (A)
3. (A) is True but (R) is False
4. Both (A) and (R) are False

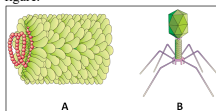
149. Which of the following features is not present in the phylum–Arthropoda?

- Metameric segmentation
- Parapodia
- Jointed appendages
- Chitinous exoskeleton

150. Which one of the following diagrams represents the placentation in *Dianthus*?



151. Identify the correct statement regarding the given figure:



1. A is a plant pathogen and has single stranded RNA as its genetic material; B infects bacteria and usually has double stranded DNA as its genetic material.
2. A is an animal pathogen and has double stranded RNA as its genetic material; B infects bacteria and usually has double stranded DNA as its genetic material.
3. A infects bacteria and usually has double stranded DNA as its genetic material; B is a plant pathogen and has single stranded RNA as its genetic material.
4. A is a plant pathogen and has single stranded RNA as its genetic material; B infects bacteria and usually has single stranded RNA as its genetic material.

152. Which of the following components provides sticky character to the bacterial cell?

1. Nuclear membrane
2. Plasma membrane
3. Glycocalyx
4. Cell Wall

153. The correctly written binomial epithet of Mango is

1. *Mangifera indica* linn
2. *Mangifera indica* Linn.
3. *Mangifera Indica* L
4. *Mangifera indica* L

154. Match each item in Column I with one in Column II and select the correct match from the codes given:

Column I [Placentation]	Column II [Example]
A Axile	P Mustard
B Parietal	Q <i>Dianthus</i>
C Free central	R Lemon
D Basal	S Marigold

Codes:

	A	B	C	D
1.	R	P	Q	S
2.	P	R	Q	S
3.	P	Q	R	S
4.	R	P	S	Q

155. The difference between mitosis anaphase and anaphase I of meiosis is:

1.	Chromosomes line up at the equatorial plate in anaphase I of meiosis
2.	Sister chromatids separate in mitosis anaphase, whereas homologous chromosomes separate in anaphase I of meiosis
3.	Only mitosis anaphase leads to genetically identical daughter cells
4.	Anaphase I of meiosis does not involve spindle fibers

156. Match List-I with List-II:

List-I	List-II
(a) Adhering junctions	(i) Establish a barrier that prevents leakage of extracellular things across a layer of cells
(b) Tight junctions	(ii) Functions like rivets and fasten cells together into strong sheets
(c) Gap junctions	(iii) Pass information through neurotransmitters from one cell to another
(d) Synaptic junctions	(iv) Provide cytoplasmic channels from one cell to an adjacent cell for communication

Choose the correct answer from the options given below

	(a)	(b)	(c)	(d)
1.	(iv)	(iii)	(ii)	(i)
2.	(i)	(iii)	(ii)	(iv)
3.	(ii)	(i)	(iv)	(iii)
4.	(i)	(ii)	(iii)	(iv)

157. Unlike roundworms, flatworms:

1. are triploblastic
2. exhibit bilateral symmetry
3. do not have a coelom
4. reproduce only by sexual methods

158. What does the term 'stele' refer to in plant anatomy?

1.	The center of the stem or root containing xylem, phloem, and pith
2.	The outer protective layer of the stem or root
3.	The arrangement of vascular bundles in the leaf
4.	The terminal part of the root that absorbs water

159. Identify the incorrect comparison between monocots and dicots in general:

Character	Monocots	Dicots
1. Leaves	Parallel venation	Reticulate venation
2. Roots	Primary root of short duration, replaced by adventitious roots forming fibrous or fleshy root systems	Develops from the radicle. Primary root often persists forming strong tap roots and secondary roots
3. Plant stem: Vascular bundles	Ring of primary bundles with cambium, differentiated into cortex and stele	Numerous scattered bundles in ground parenchyma, cambium mostly absent, no differentiation between cortical and stelar regions
4. Flowers	Parts in threes (trimerous) or multiples of three	Fours (tetramerous) or fives (pentamerous)

160. Consider the given two statements:

Assertion (A):	Lipids separate in the macromolecular fraction of a cell.
Reason (R):	Lipids are associated with membranes in a cell.

1.	Both (A) and (R) are True but (R) does not explain (A).
2.	(A) is True but (R) is False.
3.	Both (A) and (R) are True and (R) correctly explains the (A).
4.	(A) is False but (R) is True.

161. Consider the given two statements:

Assertion (A):	Mitosis is essential for the growth and repair of multicellular organisms.
Reason (R):	Mitosis usually results in the production of diploid daughter cells with identical genetic complement, which helps in restoring the nucleo-cytoplasmic ratio.

1.	Both (A) and (R) are True and (R) correctly explains (A)
2.	Both (A) and (R) are True but (R) does not correctly explain (A)
3.	(A) is True but (R) is False
4.	Both (A) and (R) are False

162. Mesorchium in frogs:

1.	is a double fold of peritoneum that adheres testes to kidney.
2.	is a double fold of peritoneum that adheres ovaries to kidney.
3.	is the outer cortical part of testes.
4.	is the outer cortical part of ovaries.

163. As we go lower from kingdom to species, the number of common characters at each taxon:

- Goes on decreasing
- Goes on increasing
- Remains same
- There is no pattern to this

164. Wheat comes under which family

- Anacardiaceae
- Poaceae
- Muscidae
- Poals

165. Which of the following protists are the chief producers in oceans?

- Diatoms
- Dinoflagellates
- Euglenoids
- Blue green algae

166. Which of the following biomolecules has the most C-H bonds [Assume all have the same number of carbon atoms in their structure]?

1. an unsaturated fatty acid
2. a polyunsaturated fatty acid
3. a monosaccharide
4. a saturated fatty acid

167. Consider the two statements:

Assertion (A):	Prop and stilt roots are aerial in origin yet they are called roots.
Reason (R):	These roots are modified to provide mechanical support

1. Both (A) and (R) are True but (R) does not correctly explain the (A).
2. (A) is True but (R) is False.
3. Both (A) and (R) are True and (R) correctly explains the (A).
4. (A) is False but (R) is True.

168. Which of the following statements is incorrect about the epidermal tissue system in plants?

1. It forms a single-layered outermost covering, often with a waxy cuticle to reduce water loss.
2. Trichomes on the shoot are usually multicellular and may be glandular or non-glandular.
3. The stomatal apparatus consists of guard cells, stomatal pores, and surrounding subsidiary cells for regulating transpiration and gaseous exchange.
4. Root hairs arising from the epidermis are multicellular and enhance mineral absorption.

169. Which of the following statements about plasmids in prokaryotic cells is correct?

1. Plasmids are essential for the survival of bacteria under normal conditions.
2. Plasmids encode proteins that aid in photosynthesis.
3. Plasmids carry non-essential genes that confer advantages such as antibiotic resistance.
4. Plasmids are linear DNA molecules found in the cytoplasm.

170. indica is-

1. Specific epithet of mango
2. Specific epithet of Indus valley species
3. Specific epithet of wheat (indian species)
4. Specific epithet of Rice (indian species)

171. In 1969, who proposed the five kingdom classification for living organisms?

1. R.H. Whittaker
2. C. Linnaeus
3. A. Roxberg
4. Virchow

172. Consider the given two statements:

Statement I:	In roots, the protoxylem lies towards the centre (pith) and the metaxylem lies towards the periphery of the organ.
Statement II:	In stems, the protoxylem lies towards periphery and metaxylem lies towards the centre.

1. **Statement I** is incorrect; **Statement II** is incorrect
2. **Statement I** is correct; **Statement II** is correct
3. **Statement I** is incorrect; **Statement II** is correct
4. **Statement I** is correct; **Statement II** is incorrect

173. What would not be true for cyclostomes?

1. All are endoparasites on some fishes.
2. They have a circular mouth without jaw.
3. Body is devoid of scales and paired fins.
4. Circulation is of closed type.

174. Gametes in *Spirogyra* are:

1. flagellated and similar in size
2. flagellated and dissimilar in size
3. non-flagellated and dissimilar in size
4. non-flagellated and similar in size

175. Which of the following differentiates bony fishes from the cartilaginous fishes?

A:	Presence of a swim bladder in bony fishes.
B:	A gill cover or operculum in cartilaginous fishes.

- Only **A** is true.
- Only **B** is correct.
- Both **A** and **B** are correct.
- Both **A** and **B** are incorrect.

176. Centrosomes are involved in:

- ATP synthesis
- Formation of spindle fibres during cell division
- Photosynthesis
- Protein modification

177. Consider the given two statements:

Statement I:	Eukaryotic cells possess an organised nucleus with a nuclear envelope and extensive compartmentalisation of cytoplasm through the presence of membrane-bound organelles.
Statement II:	All eukaryotic cells are characterized by the presence of cell wall.

- Statement I** is correct; **Statement II** is correct
- Statement I** is correct; **Statement II** is incorrect
- Statement I** is incorrect; **Statement II** is correct
- Statement I** is incorrect; **Statement II** is incorrect

178. What is the name of the type of seed where the endosperm is not completely used up during seed development, and remains in the mature seed, as seen in Castor?

1. Exalbuminous seed	2. Non-endospermic seed
3. Albuminous seed	4. Perispermic seed

179. Match the following cell structure with its characteristic feature:

Column I	Column II
(a) Tight junctions	(i) Cement neighboring cells together to form a sheet
(b) Adhering junctions	(ii) Transmit information through chemicals to another cell
(c) Gap junctions	(iii) Establish a barrier to prevent leakage of fluid across epithelial cells
(d) Synaptic junctions	(iv) Cytoplasmic channels to facilitate communication between adjacent cells

Select the correct option from the following:

Options:	(a)	(b)	(c)	(d)
1.	(ii)	(iv)	(i)	(iii)
2.	(iv)	(ii)	(i)	(iii)
3.	(iii)	(i)	(iv)	(ii)
4.	(iv)	(iii)	(i)	(ii)

180. Epigynous flowers are not seen in:

1. Ray florets of sunflower	2. Guava
3. Cucumber	4. Mustard



Mitoprep

Sapno Ki Udaan